

Supplemental Material

Inspectable Skill Matching for Accountable Candidate Ranking

This document collects the material that does not fit in the paper, in the paper’s order: Section 1 complements Sect. 5 (knowledge-based skill linking and matching), Section 2 Sect. 6 (experimental design) and Section 3 Sect. 8 (results). Tables and figures are numbered S1, S2, ... to keep them apart from those of the paper.

1 Knowledge-based Skill Linking and Matching

1.1 Skill linking

Table S1 summarises how the skill mentions of the jobs and of the CVs are linked to ESCO and to O*NET: how many are exact matches of a KB label or alias, how many are resolved by the LLM re-ranking of the retrieved candidates, and how many find no match, with the confidence of the linker and the semantic score of the selected skill. Figure S1 shows the distribution of that score for the mentions used in the experiments.

Table S1: Skill linking (normalization) of the job and CV skill mentions to ESCO and O*NET. Decision: exact match against a KB label or alias, LLM re-ranking of the retrieved candidates, or no match. Confidence: as returned by the linker. Semantic score: similarity between the mention and the selected KB skill (mean $\pm$ sd).

Side	KB	Mentions	Decision (%)			Confidence (%)			Sem. score
			Exact	Rerank	None	High	Med.	Low	
<i>All mapped skill mentions</i>									
Jobs	ESCO	50,531	18.2	81.3	0.4	95.7	0.0	4.2	.800 ± .157
Jobs	O*NET	50,531	4.6	94.1	1.3	91.8	0.0	8.2	.688 ± .167
CVs	ESCO	11,750	18.8	80.9	0.3	97.8	0.0	2.2	.822 ± .145
CVs	O*NET	11,750	7.1	92.5	0.4	94.3	0.0	5.7	.702 ± .168
<i>Mentions of the 254 jobs and 737 CVs used in the experiments</i>									
Jobs	ESCO	1,273	15.0	84.4	0.6	95.3	0.1	4.6	.787 ± .163
Jobs	O*NET	1,273	3.5	94.9	1.6	91.4	0.0	8.6	.675 ± .164
CVs	ESCO	11,694	18.8	80.9	0.3	97.8	0.0	2.1	.822 ± .145
CVs	O*NET	11,694	7.1	92.5	0.4	94.4	0.0	5.6	.702 ± .168

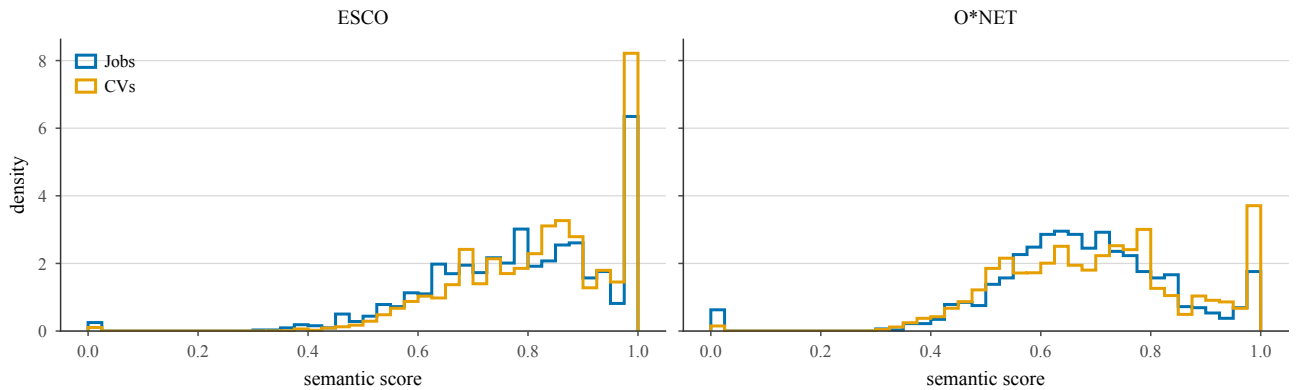


Figure S1: Semantic score of the linked skill mentions of the jobs and CVs used in the experiments, per KB.

1.2 A worked example of KB-based skill matching

Job 3031 (a nurse position) and CV 447 (a health professional) share no ESCO skill, so exact matching (T_{SY}) scores the pair 0. The KB arms still score it, and the evidence is readable. For sim_{ne} the bridge is a set of occupations: eight ESCO occupations require both *geriatrics* and *orthopaedics*. For sim_{hop} the bridge is

the taxonomy: *geriatrics* and *orthopaedics* are two steps apart under *medicine*, whereas *numerology* has to climb to *knowledge* and back, eight steps, past the decay cut-off, and scores 0.

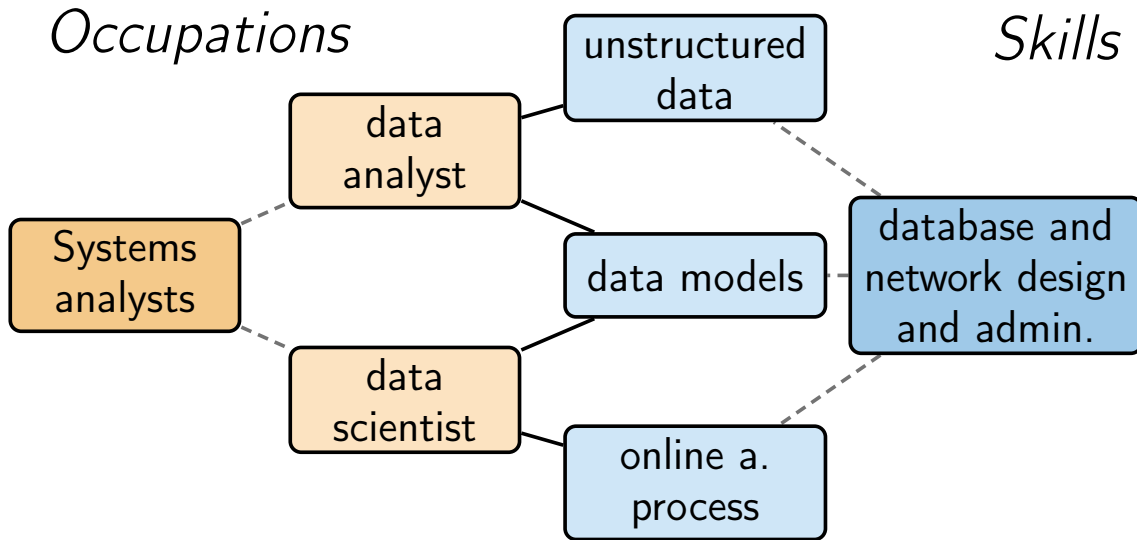


Figure S2: The ESCO fragment around the job and the CV of the worked example.

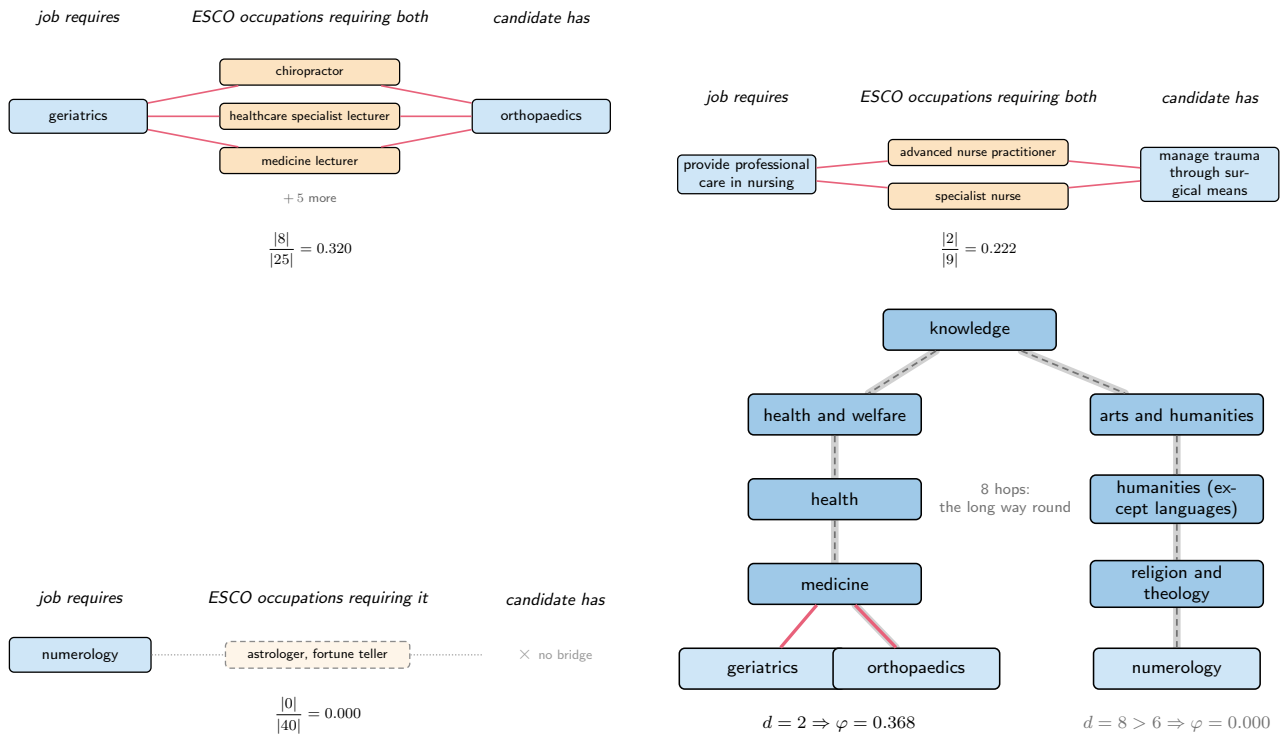


Figure S3: The evidence behind the KB scores of the worked example: one panel per bridged requirement, one for the requirement with no bridge, and one for the taxonomy.

Table S2: Scores of the worked example: for each required skill of the job, the best-matching skill of the CV, the occupations the two share, and the resulting sim_{ne} and sim_{hop} scores.

Required skill (job)	Best candidate skill	Shared occ.	s_{ne}	s_{hop}
geriatrics	orthopaedics	8/25	0.320	0.368
numerology	<i>none</i>	0/40	0.000	0.000
provide professional care in nursing	manage trauma through surgical means	2/9	0.222	0.135
Mean over required skills			0.181	0.168
Exact ESCO match (T_{SY}^{esco})			0.000	

2 Experimental Design

2.1 Relevance judgment

The prompt given to every LLM judge; the judge never sees the KB.

System prompt.

You are an expert technical recruiter deciding whether a candidate should be suitable for a job offer. You will receive a JOB OFFER (occupation, years of experience required, education level, required skills, required languages, description) and a CANDIDATE CV (years of experience, education level, languages, skills).
IMPORTANT: do NOT use ESCO (European Skills, Competences, Qualifications and Occupations) nor O*NET (Occupational Information Network). Do not map occupations or skills to ESCO or O*NET codes, concepts, groups or hierarchies, and do not use ESCO or O*NET skill-occupation relations to decide the match. Judge only from the text of the job offer and the CV, as a human recruiter would.
Evaluate the match by considering, in order of importance:
1. **SKILL FIT:** compare skills by meaning, not wording. Two skills match if a recruiter would consider them the same competence or close substitutes (e.g., different tools or terms within the same specialty). Do not reward word overlap between unrelated competencies, and do not penalize different wording for the same competence.
2. **EXPERIENCE, EDUCATION, LANGUAGES:** they can lower the score when clearly insufficient for the job’s requirements. Education is on the EQF scale: 1-4 basic/secondary, 5 short-cycle higher, 6 Bachelor’s, 7 Master’s, 8 Doctorate.
Assign exactly one score:
0 = Bad fit : the candidate’s field does not fit the job’s occupation; a recruiter would discard this CV immediately.
1 = Right or adjacent fit, but with clear gaps in required skills, experience, or education.
2 = Fit: the field matches and the requirements are essentially met, with at most minor gap.
Respond with a single digit: 0, 1, or 2. Nothing else.

User message. Each field holds the raw value of the job or of the CV; lists are comma-separated and an empty field reads **Not specified**.

JOB OFFER:
Occupation: <raw_occupation>
Years of experience required: <years_experience>
Education (EQF level): <edu_eqf>
Required skills: <raw_skills>
Required languages: <languages>
Job description: <eng_description>

CANDIDATE CV:
Years of experience: <years_experience>
Education (EQF level): <edu_eqf>
Languages: <languages>
Skills: <raw_skills>

Relevance score:

Sampling. Each pair is judged $k = 5$ times at temperature $T = 0.6$ and labelled with the median of the k runs. Judges: *qwen3:8b*, *qwen3:14b*, *ministral-3:14b*.

2.2 Ranker

Table S3: Hyper-parameters of the LambdaMART ranker, the same for every arm and every judge.

Setting	Value
Library	LightGBM 4.6.0, LGBMRanker (LambdaMART)
Objective	lambdarank on the graded labels $y \in \{0, 1, 2\}$
Grid	num_leaves $\in \{15, 31\}$; learning_rate $\in \{0.05, 0.1\}$; n_estimators $\in \{100, 300\}$
Model selection	highest mean $nDCG@10$ on the validation queries, per fold and per arm
Fixed	objective = lambdarank, random_state = 42, force_row_wise = True, n_jobs = -1, verbose = -1
Defaults	boosting_type = gbd, colsample_bytree = 1.0, importance_type = split, max_depth = -1, min_child_samples = 20, min_child_weight = 0.001, min_split_gain = 0.0, reg_alpha = 0.0, reg_lambda = 0.0, subsample = 1.0, subsample_for_bin = 200000, subsample_freq = 0
Cross-validation	5 outer folds, disjoint in the jobs; inside each, one 75/25 train/validation split of the remaining jobs (seed 42)
Placebo	$R = 80$ permutations of the sim_{hop} scores, each a full cross-validated run

2.3 Job sectors

Table S4: Job sectors after the manual merge of the ESCO occupation groups (no longer ESCO-compliant, see Sect. 6). Jobs: offers in the dataset after preprocessing (9,719 before it); Queries: jobs used as queries in the application simulation; CVs: candidates of the sector after preprocessing (742 before it); Pairs: job-CV pairs whose job belongs to the sector.

Job sector	Jobs	Queries	CVs	Pairs	Merged ESCO occupation groups
Business and administration	2,841	41	125	2,977	Administrative and commercial managers; Business and administration associate professionals; Business and administration professionals; Chief executives, senior officials and legislators
ICT	1,774	29	89	2,153	Information and communications technicians; Information and communications technology professionals
Science and engineering	802	30	94	2,261	Science and engineering associate professionals; Science and engineering professionals
Skilled trades	779	12	19	955	Building and related trades workers, excluding electricians; Electrical and electronic trades workers; Food processing, wood working, garment and other craft and related trades workers; Handicraft and printing workers; Metal, machinery and related trades workers
Management (production and services)	758	12	31	873	Hospitality, retail and other services managers; Production and specialised services managers
Sales and customer service	756	19	59	1,434	Customer services clerks; Sales workers; Street and related sales and service workers
Elementary and manual labour	527	16	50	1,258	Agricultural, forestry and fishery labourers; Cleaners and helpers; Food preparation assistants; Labourers in mining, construction, manufacturing and transport; Refuse workers and other elementary workers
Health and care	407	13	40	960	Health associate professionals; Health professionals; Personal care workers
Personal and protective services	334	15	45	1,116	Personal service workers; Protective services workers
Clerical support	185	34	104	2,648	General and keyboard clerks; Numerical and material recording clerks; Other clerical support workers
Legal, social and cultural	163	21	66	1,566	Legal, social and cultural professionals; Legal, social, cultural and related associate professionals
Education	87	12	15	915	Teaching professionals
Total	9,413	254	737	19,116	

3 Results

3.1 LLM judges and ground truth

Table S5: LLM-judged ground truth over the job–CV pairs. For each relevance grade y : *all* is the share of all pairs labelled y , *suit.* the share of the grade- y pairs whose CV is suitable (same job sector as the job). Self-consistency κ : share of pairs whose median label is backed by that fraction of the $k = 5$ runs. In bold the judge of the main results; mean and sd are across the 3 judges.

Model	Label distribution (% of pairs)						Self-consistency κ (% of pairs)				
	$y = 0$		$y = 1$		$y = 2$		.2	.4	.6	.8	1.0
	all	suit.	all	suit.	all	suit.					
qwen3:8b	74.4	40.9	18.0	58.5	7.6	69.5	.03	.09	3.40	4.45	92.04
qwen3:14b	65.0	39.4	27.2	56.5	7.7	67.7	.00	.00	2.23	2.75	95.02
ministral-3:14b	79.6	41.9	12.9	59.7	7.5	69.0	.16	.59	10.51	10.17	78.56
mean	73.0	40.7	19.4	58.2	7.6	68.8	.06	.22	5.38	5.79	88.54
sd	7.4	1.3	7.3	1.6	.1	.9	.09	.32	4.48	3.89	8.77

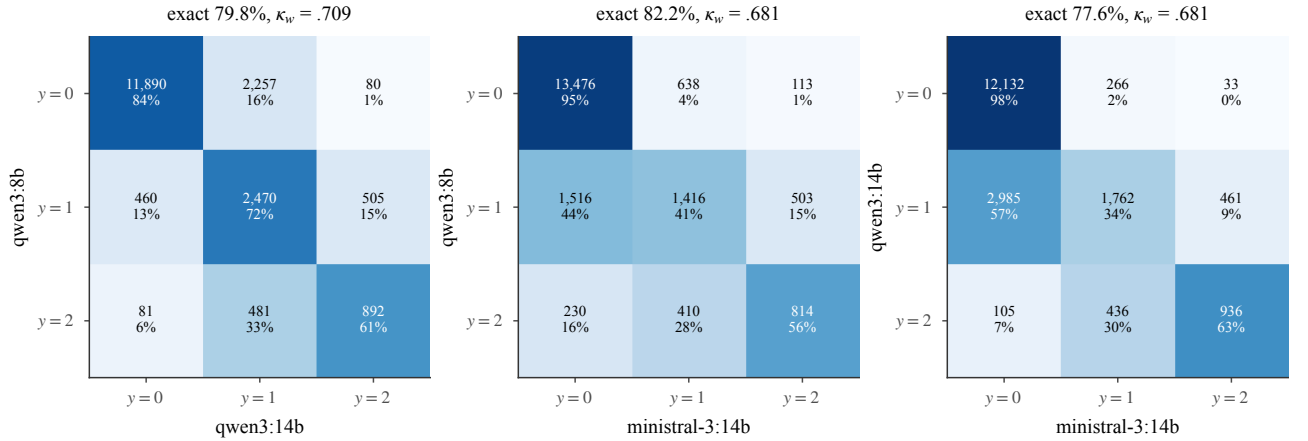


Figure S4: Confusion matrices of the labels of each pair of judges over the same job–CV pairs (count and row share).

Table S6: Agreement between the LLM judges on the label y of the same job–CV pairs. κ_w : Cohen’s kappa with quadratic weights; Up/Down: share of the pairs that the second judge labels higher/lower than the first; $0 \leftrightarrow 2$: pairs labelled 0 by one judge and 2 by the other.

Judges	Exact (%)	κ_w	κ	Up (%)	Down (%)	$0 \leftrightarrow 2$
qwen3:8b vs. qwen3:14b	79.8	.709	.562	14.9	5.3	161
qwen3:8b vs. ministral-3:14b	82.2	.681	.529	6.6	11.3	343
qwen3:14b vs. ministral-3:14b	77.6	.681	.492	4.0	18.4	138

3.2 nDCG@10

Table S7: $nDCG@10$ per judge and KB: mean over the judge’s evaluable queries $\pm$ the half-width of its 95% CI (t-interval). Best arm of each judge and KB in **bold**. T_{BN} , T_{PL} , T_{RW} and $T_{TF}^{(r)}$ do not depend on the KB and are shown once per judge, centred over its two rows.

Judge	KB	T_{BN}	T_{PL}	T_{RW}	T_{SY}	$T_{KB}^{(ne)}$	$T_{KB}^{(hop)}$	$T_{TF}^{(n)}$	$T_{TF}^{(r)}$
qwen3:8b	ESCO	.275 $\pm$.022	.248 $\pm$.019	.432 $\pm$.034	.538 $\pm$.033	.577 $\pm$.030	.593 $\pm$.031	.626 $\pm$.030	.673 $\pm$.028
	O*NET				.498 $\pm$.032	.548 $\pm$.031	.575 $\pm$.031	.604 $\pm$.031	
qwen3:14b	ESCO	.318 $\pm$.023	.306 $\pm$.022	.505 $\pm$.035	.618 $\pm$.032	.645 $\pm$.032	.660 $\pm$.031	.694 $\pm$.028	.757 $\pm$.025
	O*NET				.568 $\pm$.032	.613 $\pm$.031	.632 $\pm$.031	.673 $\pm$.028	
ministral-3:14b	ESCO	.245 $\pm$.022	.219 $\pm$.018	.412 $\pm$.036	.513 $\pm$.035	.559 $\pm$.034	.563 $\pm$.035	.603 $\pm$.034	.653 $\pm$.032
	O*NET				.488 $\pm$.035	.520 $\pm$.035	.544 $\pm$.034	.596 $\pm$.034	

Table S8: $nDCG@10$ per KB: mean $\pm$ sd, over the 3 LLM judges, of each judge’s mean $nDCG@10$. Best arm of each KB in **bold**; T_{BN} , T_{PL} , T_{RW} and $T_{TF}^{(r)}$ do not depend on the KB and are shown once, centred over the two rows. Each judge’s value is computed over its own evaluable queries and its own labels, so the sd reflects the judges’ different scales as well as the arms’ behaviour; absolute $nDCG@10$ levels are not directly comparable across judges.

KB	T_{BN}	T_{PL}	T_{RW}	T_{SY}	$T_{KB}^{(ne)}$	$T_{KB}^{(hop)}$	$T_{TF}^{(n)}$	$T_{TF}^{(r)}$
ESCO	.279 $\pm$.036	.258 $\pm$.044	.450 $\pm$.049	.557 $\pm$.055	.593 $\pm$.045	.605 $\pm$.050	.641 $\pm$.047	.694 $\pm$.055
O*NET				.518 $\pm$.044	.560 $\pm$.048	.584 $\pm$.044	.624 $\pm$.043	

Table S9: Arm summary for the judge *qwen3:8b*. $nDCG@10$, the gain Γ over T_{BN} and its 95% CI are computed over the judge’s 245 evaluable queries (at least one relevant candidate); Ties and $\overline{SPR}_y$ over all the queries holding a grade- y candidate. ($\star$) marks a difference from T_{BN} significant after Holm–Bonferroni over the 66 pairs of arms of both KBs; for T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule). Ties: % of tied candidate pairs in the ranking; $\overline{SPR}_y$: mean Percentile Ranking of the grade- y candidates (lower is closer to the top). Best in **bold**.

Arm	KB	Norm.	γ	$nDCG@10$	Γ	CI	Ties	$\overline{SPR}_0$	$\overline{SPR}_1$	$\overline{SPR}_2$
T_{BN}	–	–	–	.275	–	–	37.2	.519	.458	.416
T_{PL}	–	✓	perm.	.248	–.027	–	.3	.514	.475	.423
T_{RW}	–	✗	1	.432	.157*	[.131, .183]	32.1	.544	.418	.327
T_{SY}	ESCO	✓	1	.538	.263*	[.233, .294]	28.8	.553	.377	.253
$T_{KB}^{(ne)}$	ESCO	✓	sim_{ne}	.577	.302*	[.272, .331]	2.0	.570	.317	.184
$T_{KB}^{(hop)}$	ESCO	✓	sim_{hop}	.593	.318*	[.287, .348]	2.7	.567	.328	.183
$T_{TF}^{(n)}$	ESCO	✓	cos	.626	.352*	[.322, .381]	.3	.577	.304	.155
T_{SY}	O*NET	✓	1	.498	.223*	[.194, .251]	28.5	.551	.393	.263
$T_{KB}^{(ne)}$	O*NET	✓	sim_{ne}	.548	.273*	[.243, .302]	1.4	.563	.343	.207
$T_{KB}^{(hop)}$	O*NET	✓	sim_{hop}	.575	.300*	[.270, .331]	5.3	.568	.333	.194
$T_{TF}^{(n)}$	O*NET	✓	cos	.604	.329*	[.298, .359]	.3	.573	.322	.163
$T_{TF}^{(r)}$	–	✗	cos	.673	.398*	[.369, .428]	.5	.579	.301	.151

Table S10: Arm summary for the judge *qwen3:14b*. $nDCG@10$, the gain Γ over T_{BN} and its 95% CI are computed over the judge’s 247 evaluable queries (at least one relevant candidate); Ties and $\overline{SPR}_y$ over all the queries holding a grade- y candidate. (★) marks a difference from T_{BN} significant after Holm–Bonferroni over the 66 pairs of arms of both KBs; for T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule). Ties: % of tied candidate pairs in the ranking; $\overline{SPR}_y$: mean Percentile Ranking of the grade- y candidates (lower is closer to the top). Best in **bold**.

Arm	KB	Norm.	γ	$nDCG@10$	Γ	CI	Ties	$\overline{SPR}_0$	$\overline{SPR}_1$	$\overline{SPR}_2$
T_{BN}	–	–	–	.318	–	–	37.2	.520	.471	.421
T_{PL}	–	✓	perm.	.306	–.012	–	.3	.516	.482	.429
T_{RW}	–	✗	1	.505	.187*	[.159, .215]	32.1	.560	.420	.323
T_{SY}	ESCO	✓	1	.618	.300*	[.271, .330]	28.8	.573	.396	.218
$T_{KB}^{(ne)}$	ESCO	✓	sim_{ne}	.645	.327*	[.297, .357]	2.0	.592	.352	.175
$T_{KB}^{(hop)}$	ESCO	✓	sim_{hop}	.660	.343*	[.313, .372]	2.7	.588	.360	.169
$T_{TF}^{(n)}$	ESCO	✓	cos	.694	.376*	[.348, .404]	.3	.602	.343	.140
T_{SY}	O*NET	✓	1	.568	.250*	[.222, .278]	28.5	.568	.405	.253
$T_{KB}^{(ne)}$	O*NET	✓	sim_{ne}	.613	.295*	[.266, .325]	1.4	.584	.354	.188
$T_{KB}^{(hop)}$	O*NET	✓	sim_{hop}	.632	.314*	[.285, .343]	5.3	.587	.370	.185
$T_{TF}^{(n)}$	O*NET	✓	cos	.673	.355*	[.327, .384]	.3	.597	.346	.151
$T_{TF}^{(r)}$	–	✗	cos	.757	.439*	[.412, .466]	.5	.606	.330	.127

Table S11: Arm summary for the judge *ministral-3:14b*. $nDCG@10$, the gain Γ over T_{BN} and its 95% CI are computed over the judge’s 235 evaluable queries (at least one relevant candidate); Ties and $\overline{SPR}_y$ over all the queries holding a grade- y candidate. (★) marks a difference from T_{BN} significant after Holm–Bonferroni over the 66 pairs of arms of both KBs; for T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule). Ties: % of tied candidate pairs in the ranking; $\overline{SPR}_y$: mean Percentile Ranking of the grade- y candidates (lower is closer to the top). Best in **bold**.

Arm	KB	Norm.	γ	$nDCG@10$	Γ	CI	Ties	$\overline{SPR}_0$	$\overline{SPR}_1$	$\overline{SPR}_2$
T_{BN}	–	–	–	.245	–	–	37.2	.514	.476	.370
T_{PL}	–	✓	perm.	.219	–.027	–	.3	.511	.482	.398
T_{RW}	–	✗	1	.412	.167*	[.139, .195]	32.1	.535	.418	.293
T_{SY}	ESCO	✓	1	.513	.268*	[.235, .300]	28.8	.542	.381	.231
$T_{KB}^{(ne)}$	ESCO	✓	sim_{ne}	.559	.314*	[.278, .349]	2.0	.552	.327	.178
$T_{KB}^{(hop)}$	ESCO	✓	sim_{hop}	.563	.318*	[.283, .353]	2.7	.549	.333	.191
$T_{TF}^{(n)}$	ESCO	✓	cos	.603	.358*	[.323, .392]	.3	.557	.297	.155
T_{SY}	O*NET	✓	1	.488	.243*	[.210, .275]	28.5	.540	.379	.251
$T_{KB}^{(ne)}$	O*NET	✓	sim_{ne}	.520	.274*	[.238, .310]	1.4	.544	.343	.212
$T_{KB}^{(hop)}$	O*NET	✓	sim_{hop}	.544	.299*	[.265, .333]	5.3	.551	.332	.194
$T_{TF}^{(n)}$	O*NET	✓	cos	.596	.351*	[.316, .386]	.3	.554	.315	.172
$T_{TF}^{(r)}$	–	✗	cos	.653	.408*	[.373, .443]	.5	.557	.294	.154

Table S12: Arm summary averaged over the 3 LLM judges: mean $\pm$ sd across judges of $nDCG@10$, of the gain Γ over T_{BN} and of $\overline{SPR}_y$. (★): the difference from T_{BN} is significant for every judge (Holm–Bonferroni within each judge). Ties depend on the features only and are the same for every judge. Best in **bold**. Each judge’s value is computed over its own evaluable queries and its own labels, so the sd reflects the judges’ different scales as well as the arms’ behaviour; absolute $nDCG@10$ levels are not directly comparable across judges.

Arm	KB	Norm.	γ	$nDCG@10$	Γ	Ties	$\overline{SPR}_0$	$\overline{SPR}_1$	$\overline{SPR}_2$
T_{BN}	–	–	–	.279 $\pm$.036	–	37.2	.518 $\pm$.003	.468 $\pm$.009	.402 $\pm$.028
T_{PL}	–	✓	perm.	.258 $\pm$.044	–.022 $\pm$.008	.3	.514 $\pm$.003	.480 $\pm$.004	.417 $\pm$.016
T_{RW}	–	✗	1	.450 $\pm$.049	.170 $\pm$.015*	32.1	.546 $\pm$.012	.419 $\pm$.001	.314 $\pm$.018
T_{SY}	ESCO	✓	1	.557 $\pm$.055	.277 $\pm$.020*	28.8	.556 $\pm$.016	.385 $\pm$.010	.234 $\pm$.018
$T_{KB}^{(ne)}$	ESCO	✓	sim_{ne}	.593 $\pm$.045	.314 $\pm$.013*	2.0	.571 $\pm$.020	.332 $\pm$.018	.179 $\pm$.004
$T_{KB}^{(hop)}$	ESCO	✓	sim_{hop}	.605 $\pm$.050	.326 $\pm$.014*	2.7	.568 $\pm$.020	.340 $\pm$.017	.181 $\pm$.011
$T_{TF}^{(n)}$	ESCO	✓	cos	.641 $\pm$.047	.362 $\pm$.013*	.3	.579 $\pm$.022	.315 $\pm$.024	.150 $\pm$.009
T_{SY}	O*NET	✓	1	.518 $\pm$.044	.238 $\pm$.014*	28.5	.553 $\pm$.014	.392 $\pm$.013	.255 $\pm$.006
$T_{KB}^{(ne)}$	O*NET	✓	sim_{ne}	.560 $\pm$.048	.281 $\pm$.013*	1.4	.564 $\pm$.020	.347 $\pm$.007	.203 $\pm$.013
$T_{KB}^{(hop)}$	O*NET	✓	sim_{hop}	.584 $\pm$.044	.304 $\pm$.008*	5.3	.569 $\pm$.018	.345 $\pm$.022	.191 $\pm$.005
$T_{TF}^{(n)}$	O*NET	✓	cos	.624 $\pm$.043	.345 $\pm$.014*	.3	.574 $\pm$.022	.328 $\pm$.017	.162 $\pm$.011
$T_{TF}^{(r)}$	–	✗	cos	.694 $\pm$.055	.415 $\pm$.021*	.5	.581 $\pm$.025	.308 $\pm$.019	.144 $\pm$.015

3.3 Pairwise comparisons

Table S13: Pairwise comparison of the arms for the judge *qwen3:8b*: Holm–Bonferroni-adjusted p-values of the paired sign-flip permutation test (20,000 resamples) on the per-query *nDCG@10* differences, over the 66 pairs of arms of both KBs (E: ESCO, O: O*NET); (★): $p < 0.05$. For T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule).

	T_{BN}	T_{PL}	T_{RW}	T_{SY}^E	$T_{KB}^{(ne)E}$	$T_{KB}^{(hop)E}$	$T_{TF}^{(n)E}$	T_{SY}^O	$T_{KB}^{(ne)O}$	$T_{KB}^{(hop)O}$	$T_{TF}^{(n)O}$
T_{PL}	.089										
T_{RW}	.007★	.007★									
T_{SY}^E	.007★	.007★	.007★								
$T_{KB}^{(ne)E}$	.007★	.007★	.007★	.007★							
$T_{KB}^{(hop)E}$	.007★	.007★	.007★	.007★	.742						
$T_{TF}^{(n)E}$	.007★	.007★	.007★	.007★	.007★	.015★					
T_{SY}^O	.007★	.007★	.007★	.087	.007★	.007★	.007★				
$T_{KB}^{(ne)O}$	.007★	.007★	.007★	1.000	.345	.027★	.007★	.007★			
$T_{KB}^{(hop)O}$	.007★	.007★	.007★	.111	1.000	.826	.007★	.007★	.084		
$T_{TF}^{(n)O}$	.007★	.007★	.007★	.007★	.345	1.000	.742	.007★	.007★	.032★	
$T_{TF}^{(r)}$	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★

Table S14: Pairwise comparison of the arms for the judge *qwen3:14b*: Holm–Bonferroni-adjusted p-values of the paired sign-flip permutation test (20,000 resamples) on the per-query *nDCG@10* differences, over the 66 pairs of arms of both KBs (E: ESCO, O: O*NET); (★): $p < 0.05$. For T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule).

	T_{BN}	T_{PL}	T_{RW}	T_{SY}^E	$T_{KB}^{(ne)E}$	$T_{KB}^{(hop)E}$	$T_{TF}^{(n)E}$	T_{SY}^O	$T_{KB}^{(ne)O}$	$T_{KB}^{(hop)O}$	$T_{TF}^{(n)O}$
T_{PL}	1.000										
T_{RW}	.007★	.007★									
T_{SY}^E	.007★	.007★	.007★								
$T_{KB}^{(ne)E}$	.007★	.007★	.007★	.114							
$T_{KB}^{(hop)E}$	.007★	.007★	.007★	.007★	.754						
$T_{TF}^{(n)E}$	.007★	.007★	.007★	.007★	.007★	.007★					
T_{SY}^O	.007★	.007★	.007★	.007★	.007★	.007★	.007★				
$T_{KB}^{(ne)O}$	.007★	.007★	.007★	1.000	.312	.007★	.007★	.007★			
$T_{KB}^{(hop)O}$	.007★	.007★	.007★	1.000	1.000	.275	.007★	.007★	.552		
$T_{TF}^{(n)O}$	.007★	.007★	.007★	.007★	.450	1.000	.750	.007★	.007★	.007★	
$T_{TF}^{(r)}$	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★

Table S15: Pairwise comparison of the arms for the judge *ministral-3:14b*: Holm–Bonferroni-adjusted p-values of the paired sign-flip permutation test (20,000 resamples) on the per-query *nDCG@10* differences, over the 66 pairs of arms of both KBs (E: ESCO, O: O*NET); (★): $p < 0.05$. For T_{PL} the test runs inside each of its 80 replicates and the median p-value is doubled (Rüger’s rule).

	T_{BN}	T_{PL}	T_{RW}	T_{SY}^E	$T_{KB}^{(ne)E}$	$T_{KB}^{(hop)E}$	$T_{TF}^{(n)E}$	T_{SY}^O	$T_{KB}^{(ne)O}$	$T_{KB}^{(hop)O}$	$T_{TF}^{(n)O}$
T_{PL}	.216										
T_{RW}	.007★	.007★									
T_{SY}^E	.007★	.007★	.007★								
$T_{KB}^{(ne)E}$	.007★	.007★	.007★	.007★							
$T_{KB}^{(hop)E}$	.007★	.007★	.007★	.007★	1.000						
$T_{TF}^{(n)E}$	.007★	.007★	.007★	.007★	.007★	.007★					
T_{SY}^O	.007★	.007★	.007★	.752	.007★	.007★	.007★				
$T_{KB}^{(ne)O}$	.007★	.007★	.007★	1.000	.200	.071	.007★	.081			
$T_{KB}^{(hop)O}$	.007★	.007★	.007★	.330	1.000	1.000	.007★	.007★	.216		
$T_{TF}^{(n)O}$	.007★	.007★	.007★	.007★	.186	.235	1.000	.007★	.007★	.007★	
$T_{TF}^{(r)}$	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★	.007★

3.4 Concordance of the arm ordering across judges

Table S16: Concordance of the arm ordering across the LLM judges. Top: rank of each arm by mean $nDCG@10$ (1 = best) under each judge, per KB (Q3-8 = qwen3:8b, Q3-14 = qwen3:14b, MI-14 = ministral-3:14b). Bottom: Kendall's τ between the orderings of two judges.

Arm	ESCO			O*NET		
	Q3-8	Q3-14	MI-14	Q3-8	Q3-14	MI-14
T_{BN}	7	7	7	7	7	7
T_{PL}	8	8	8	8	8	8
T_{RW}	6	6	6	6	6	6
T_{SY}	5	5	5	5	5	5
$T_{KB}^{(ne)}$	4	4	4	4	4	4
$T_{KB}^{(hop)}$	3	3	3	3	3	3
$T_{TF}^{(n)}$	2	2	2	2	2	2
$T_{TF}^{(r)}$	1	1	1	1	1	1

Kendall τ	ESCO	O*NET
qwen3:8b vs. qwen3:14b	1.000	1.000
qwen3:8b vs. ministral-3:14b	1.000	1.000
qwen3:14b vs. ministral-3:14b	1.000	1.000

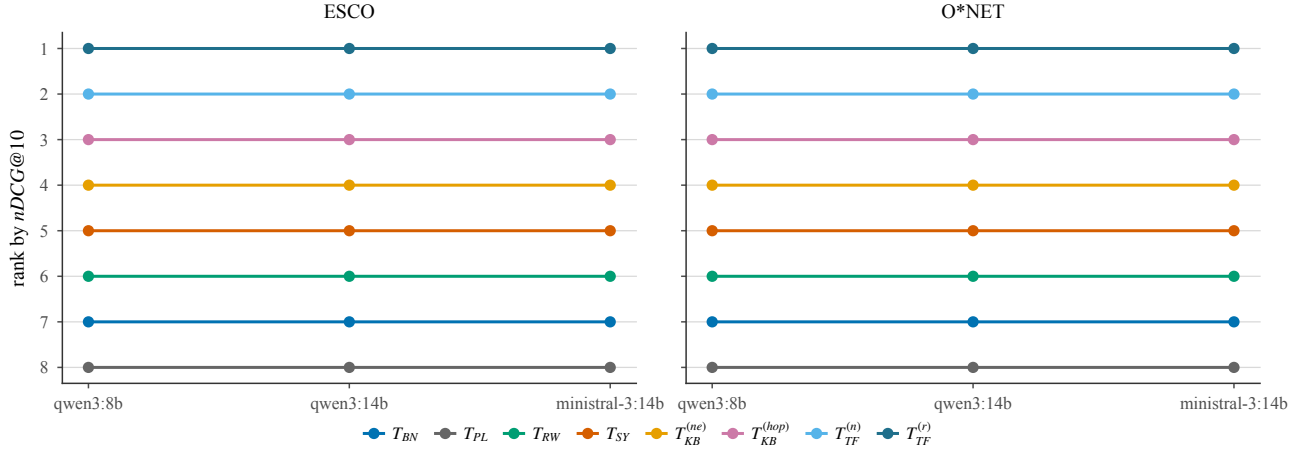


Figure S5: Rank of each arm by mean $nDCG@10$ under each judge, per KB (1 = best).

3.5 Rank Change

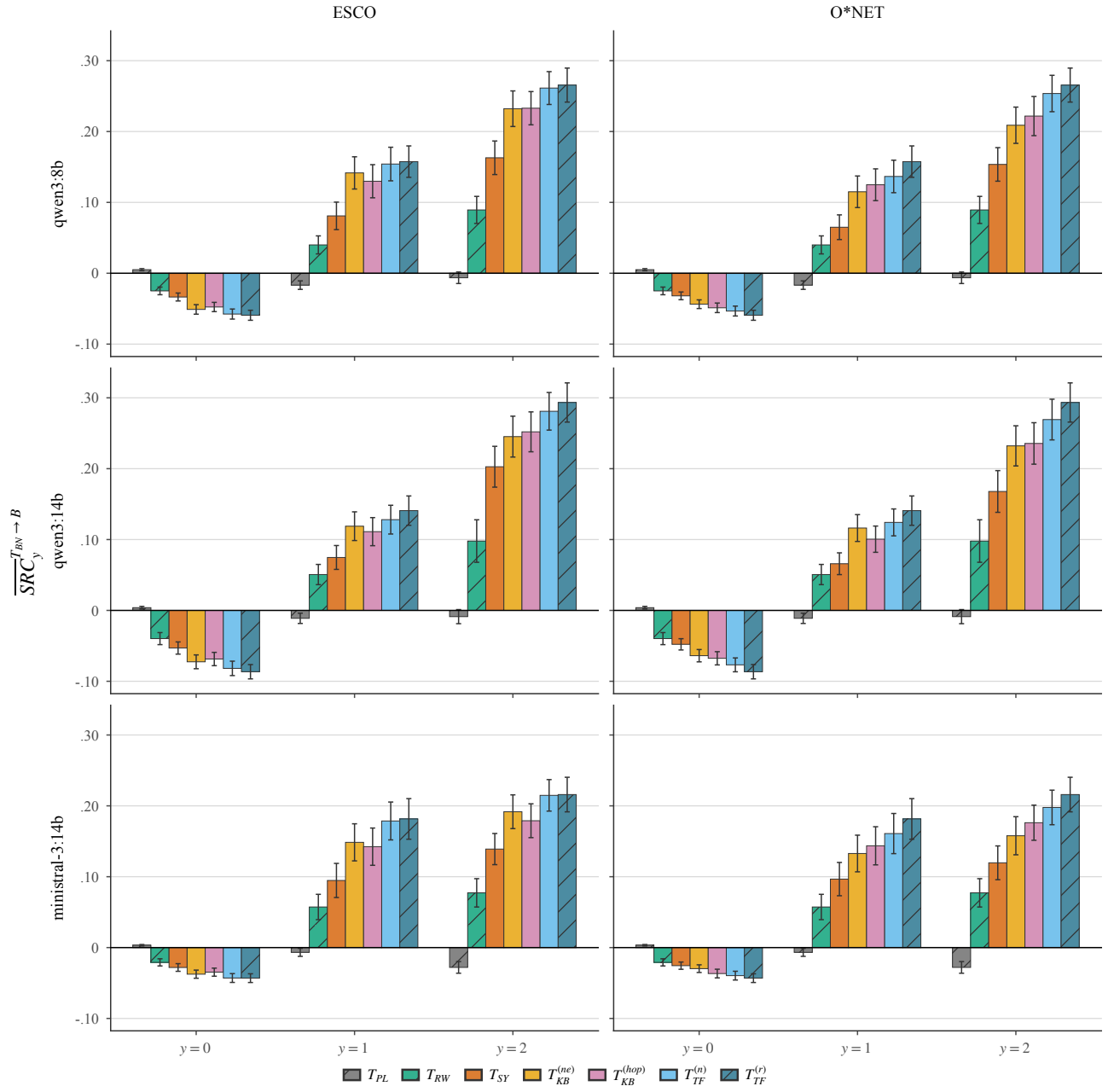


Figure S6: Stratified Rank Change $\overline{SRC}_y^{T_{BN} \rightarrow B}$ per judge (rows) and KB (columns). Error bars are 95% CIs over the jobs.

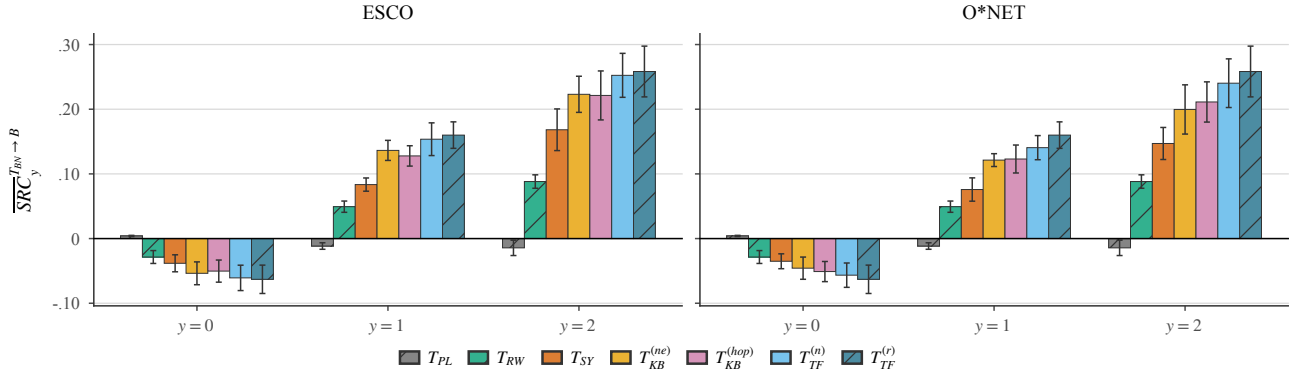


Figure S7: Stratified Rank Change $\overline{SRC}_y^{T_{BN} \rightarrow B}$ averaged over the judges, for ESCO (left) and O*NET (right). Error bars are the standard deviation across the judges.

Table S17: Stratified Rank Change $\overline{SRC}_y^{T_{BN} \rightarrow B}$ per KB and relevance grade: mean $\pm$ sd over the 3 LLM judges. A positive value means that grade- y candidates move towards the top of the ranking. Kendall's W : concordance of the judges' orderings of the 7 arms by $\overline{SRC}_y$ (1 = every judge orders them identically).

Arm	ESCO			O*NET		
	$y = 0$	$y = 1$	$y = 2$	$y = 0$	$y = 1$	$y = 2$
T_{PL}	.004 $\pm$.001	-.012 $\pm$.005	-.014 $\pm$.012	.004 $\pm$.001	-.012 $\pm$.005	-.014 $\pm$.012
T_{RW}	-.029 $\pm$.010	.049 $\pm$.009	.088 $\pm$.010	-.029 $\pm$.010	.049 $\pm$.009	.088 $\pm$.010
T_{SY}	-.038 $\pm$.013	.083 $\pm$.010	.168 $\pm$.032	-.035 $\pm$.012	.076 $\pm$.018	.147 $\pm$.025
$T_{KB}^{(ne)}$	-.054 $\pm$.018	.136 $\pm$.016	.223 $\pm$.028	-.046 $\pm$.017	.121 $\pm$.010	.200 $\pm$.038
$T_{KB}^{(hop)}$	-.050 $\pm$.017	.128 $\pm$.016	.221 $\pm$.038	-.051 $\pm$.016	.123 $\pm$.022	.211 $\pm$.031
$T_{TF}^{(n)}$	-.061 $\pm$.020	.154 $\pm$.025	.252 $\pm$.034	-.057 $\pm$.019	.141 $\pm$.019	.240 $\pm$.038
$T_{TF}^{(r)}$	-.063 $\pm$.022	.160 $\pm$.020	.258 $\pm$.039	-.063 $\pm$.022	.160 $\pm$.020	.258 $\pm$.039
Kendall's W	1.000	1.000	.984	1.000	.984	1.000